

Questions to Describe Our Project

1. What type of system are you creating?

Real estate property management

2. What is the purpose of your system?

The purpose of the system is to make managing a real estate property portfolio simpler, more organized, and less dependent on manual work. The system will allow property owners and property management staff to manage properties, units, tenants, applicants, leases, invoices, rent payments, maintenance requests, and property inspections in one centralized system.

3. Who will use the system?

The main users of the system will be:

- Property owners
- Property managers
- Accountants
- Tenants
- Maintenance staff

4. What are the required entities in your system?

The main entities in our system are:

- Property
- Unit
- Admin User
- Owner
- Tenant
- Applicant
- Lease
- Invoice
- Payment
- Work Order
- Inspection

5. What does each entity represent?

- **Property** – Represents a piece of real estate being managed, such as an apartment complex, single-family home, duplex, or commercial building.

- **Unit** – Represents an individual space within a property that can be rented, such as an apartment, house, office, or retail space.
- **Admin User** – Represents employees who work for the property management company, such as property managers and accountants. Maintenance staff may also use the system through an admin account with appropriate permissions.
- **Owner** – Represents a person who owns one or more properties managed by the system.
- **Tenant** – Represents a person who currently rents one or more units managed by the property management company.
- **Applicant** – Represents a person who has submitted an application to rent an available unit but has not yet become a tenant.
- **Lease** – Represents the rental agreement between a tenant and a unit. It contains information about the rental period and monthly rent.
- **Invoice** – Represents a bill issued to a tenant for rent or other services or charges. It contains the amount due, due date, and payment status.
- **Payment** – Represents a payment made by a tenant toward an invoice. It records information such as the amount paid, payment date, and payment method.
- **Work Order** – Represents a maintenance request submitted by a tenant to report a problem with their unit. A work order can be assigned to maintenance staff and tracked until the issue is completed.
- **Inspection** – Represents an inspection of a property or unit. Inspections can be used to document the condition of a unit, identify problems, and record inspection results.

6. What information should the system store about each entity?

1. Property:

- Property ID
- Address
- Dwelling type

- Number of units
- Owner

2. Unit:

- Unit ID
- Property ID/address
- Unit number
- Vacancy status
- Monthly rent amount

3. Admin User:

- User ID
- Name
- Phone number
- Password
- Job role
- Permissions

4. Owner:

- Owner ID
- Name
- Phone number
- Email
- Login information
- Permissions
- Properties owned

5. Tenant:

- Tenant ID
- Name
- Phone number
- Email
- Login information
- Permissions
- Unit(s) rented

6. Applicant:

- Application reference number
- Name
- Phone number
- Email
- Unit applied for
- Application date
- Application status

7. Lease:

- Lease ID
- Unit
- Tenant

- Lease start date
- Lease end date
- Monthly rent amount
- Lease document

8. Invoice:

- Invoice number
- Tenant
- Unit
- Amount due
- Due date
- Invoice status
- Invoice document

9. Payment:

- Payment ID
- Invoice
- Tenant
- Amount paid
- Payment date
- Payment method
- Payment status

10. Work Order:

- Work order number
- Tenant
- Unit
- Description of the problem
- Date submitted
- Status
- Maintenance staff assigned
- Date completed

11. Inspection:

- Inspection ID
- Property or unit
- Inspection date
- Inspector
- Inspection type
- Inspection findings
- Inspection status
- Notes

7. How are the entities connected?

The entities in the system are connected through their relationships with properties, units, tenants, and management activities:

- A property can contain one or more units.
- An owner can own one or more properties.
- Each unit belongs to one property.
- A tenant can rent one or more units over time.
- An applicant can apply for an available unit.
- A lease connects a tenant to a unit and defines the rental relationship.
- A tenant can have one or more invoices associated with their leased unit.
- An invoice can have one or more payments associated with it.
- A tenant can submit work orders for units they currently rent.
- A work order can be assigned to an admin user who is maintenance staff.
- A property or unit can have multiple inspections over time.
- An inspection can be associated with the property or specific unit being inspected.

8. What should users be able to do in the system?

The system will allow users to perform different actions based on their role and permissions.

- An admin user can add, edit, and manage properties.
- An admin user can add, edit, and manage units.
- An owner can view the properties and units they own.
- A tenant can view their lease information.
- A tenant can view their invoices and payment history.
- An admin user can create and manage invoices.
- An admin user can record and manage rent payments.
- An admin user can approve or deny an applicant.
- A tenant can submit a work order for a maintenance problem.
- Maintenance staff can view assigned work orders and update their status.
- Admin users can create and manage property or unit inspections.
- Authorized users can view inspection results and notes.

9. What rules should the system follow?

The system will follow several rules to protect information and maintain accurate property records.

- An owner can only view properties that they own.
- A tenant can only access their own lease, invoices, payments, and work orders.
- A unit cannot have more than one active lease at the same time.
- An applicant can only apply for an available unit.
- A lease must be connected to both a tenant and a unit.
- An invoice must be connected to a tenant and their respective unit.
- A payment must be connected to an invoice and tenant.
- A tenant can only create a work order for a unit they currently rent.
- A work order must have a status so its progress can be tracked.
- Only authorized admin users can approve or deny applicants.
- Only authorized users can create or modify invoices, payments, leases, and inspections.
- Admin users can only access features and make changes allowed by their permission level.
- An inspection must be associated with a property or unit and include an inspection date.

10. What is one example of how someone would use the system?

A tenant logs into the system and selects the option to create a work order. The tenant selects their unit, enters a description of the maintenance problem, and submits the request. The system creates a unique work order number and records the tenant, unit, description, submission date, and status. A maintenance staff member can then be assigned to the work order, update its status while working on the problem, and mark it as completed when the issue has been resolved.

Another example would be a tenant receiving a monthly invoice for rent. The tenant can view the amount due and due date in the system. After the rent is paid, an admin user can record the payment, connecting it to the appropriate invoice and updating the invoice status.

11. What extra entities could make the system better?

The system already includes the major entities needed to meet the requirements of a Real Estate Property Management System. However, the following additional entities could make the system more complete:

- **Notification** – Stores notifications sent to users about important events, such as upcoming rent due dates, application decisions, maintenance updates, lease expiration dates, or inspection results.
- **Property Expense** – Stores expenses associated with managing a property, such as repairs, utilities, landscaping, cleaning, or other operating costs. This could help property owners and accountants track the profitability of individual properties.

12. Final project description:

Our team selected a Real Estate Property Management System. The purpose of this system is to make managing a real estate portfolio more organized and efficient by providing one system for managing properties, units, owners, tenants, applicants, leases, invoices, rent payments, maintenance

requests, and inspections. The main users of the system are property owners, property managers, accountants, tenants, and maintenance staff.

The required entities in our system are Property, Unit, Admin User, Owner, Tenant, Applicant, Lease, Invoice, Payment, Work Order, and Inspection. These entities are important because they represent the main information and activities involved in managing rental properties. Property and Unit store information about the real estate being managed, while Owner, Tenant, Applicant, and Admin User represent the people who interact with the system. Lease, Invoice, and Payment manage the rental and financial aspects of the system, while Work Order and Inspection manage maintenance and the condition of properties and units.

The system will store information such as property addresses, dwelling types, unit numbers, vacancy status, rental prices, owner and tenant contact information, applicant information, lease dates, monthly rent amounts, invoice amounts, payment dates, maintenance requests, inspection results, and user permissions. The entities are connected because an owner can own one or more properties, and each property can contain one or more units. Applicants can apply for available units, while tenants can rent units through leases. Leases are connected to invoices and payments to keep track of rent, and tenants can submit work orders for maintenance problems in their units. Properties and units can also have inspections to record their condition and identify problems that may need to be addressed.

The system will allow users to add and manage properties and units, view properties they own, approve or deny applicants, manage tenants and leases, create and manage invoices, record rent payments, submit and complete maintenance requests, and conduct and record inspections. Access to these features will be controlled by user roles and permissions so that users can only access information and functions appropriate for their role. For example, a tenant can log into the system, select their unit, and submit a work order describing a maintenance problem. The system will create a unique work order number and record the tenant, unit, description, submission date, and status. A maintenance staff

member can then view the work order, update its status, and mark it as completed after resolving the problem.

We may also include additional entities such as Notification and Property Expense to make the system more complete. A Notification entity could provide users with updates about rent payments, maintenance requests, applications, leases, and inspections. A Property Expense entity could track expenses such as repairs, utilities, and landscaping. Overall, the system is designed to centralize property management information, improve organization, and reduce the amount of manual work required to manage a real estate portfolio.